PDKM BioTech
Chemicals & Medicine Div. of KM Group

Signature Manifestation

Batch Record Approval Signatures :

Prepared By: Arya Stark Date: 22-Aug-2023
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Approved By: Lannister Tyrion Date: 23-Aug-2023
Tyrion Lannister
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Clegane Sandor Date: 23-Aug-2023
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PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05- 9700
	Document Version Number: 2.3

1. Bill Of Material

Serial Number	Material ID	Material Description	Lot Number	Quantity	Expiry Date	Issued By	Received By
1	A2R7F9	Acetaminophen	579111	①			
2	K8J3M5	Ascorbic Acid					
3	B6N2P4	Magnesium Stearate					
4	Q1W5E7	Microcrystalline Cellulose					
5	X9C4V6	Ethanol					
6	G3T8H2	Sodium Chloride					
7	Z7Y6U1	Lactose Monohydrate					
8	D5F1R9	Polyethylene Glycol					
9	L2K6J4	Propylene Glycol					
10	M4N9B8	Hydroxypropyl Methylcellulose					
11	E7Q1W5	Croscarmellose Sodium					
12	V6X2C3	Povidone					
13	H9G4T7	Sodium Lauryl Sulfate					
14	U1Z8Y5	Dimethyl Sulfoxide (DMSO)					
15	P3D6F2	Gelatin					
16	R5A7E9	Polyvinyl Alcohol (PVA)					
17	T7H2J6	Stearic Acid					
18	Y4V8K1	Titanium Dioxide					

① wrong Lot number

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1. Bill Of Material

Serial Number	Material ID	Material Description	Lot Number	Quantity	Expiry Date	Issued By	Received By
1	A2R7F9	Acetaminophen	ST9124	90ML	09-NOV-2025		
2	K8J3M5	Ascorbic Acid	ST5621	1.2G	01-DEC-2025		
3	B6N2P4	Magnesium Stearate	ST9429	20.5L	25-JAN-2026		
4	Q1W5E7	Microcrystalline Cellulose	ST4091	3L	11-MAR-2025		
5	X9C4V6	Ethanol	ST3014	11G	04-OCT-2025		
6	G3T8H2	Sodium Chloride	ST8192	90MG	21-OCT-2025		
7	Z7Y6U1	Lactose Monohydrate	ST2108	1ML	29-JAN-2026		
8	D5F1R9	Polyethylene Glycol	ST9146	2G	17-JAN-2026		
9	L2K6J4	Propylene Glycol	ST5169	100ML	15-APR-2026		
10	M4N9B8	Hydroxypropyl Methylcellulose	ST8437	2ML	11-JUN-2025	22-Aug-23	22-Aug-23
11	E7Q1W5	Croscarmellose Sodium	ST9924	3L	04-MAY-2025		
12	V6X2C3	Povidone	ST3140	1.5G	07-APR-2025		
13	H9G4T7	Sodium Lauryl Sulfate	ST6789	12ML	25-MAR-2025		
14	U1Z8Y5	Dimethyl Sulfoxide (DMSO)	ST7692	1ML	19-FEB-2025		
15	P3D6F2	Gelatin	ST3456	2G	29-MAY-2024		
16	R5A7E9	Polyvinyl Alcohol (PVA)	ST8259	20L	24-DEC-2024		
17	T7H2J6	Stearic Acid	ST4881	140ML	15-JAN-2025		
18	Y4V8K1	Titanium Dioxide	ST3863	2L	26-OCT-2025		

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2. Equipment List

Serial Number	Equipment	Manufacturer	Equipment Number	Experiemnt Used	Initials
1	High-Performance Liquid Chromatography (HPLC) System	Thermo Fisher Scientific	HPLC-05	✓ Yes No NA Caliberation Due Date : 31-Dec-2025	
2	Gas Chromatography-Mass Spectrometry (GC-MS)	Thermo Fisher Scientific	GCMS-A11	✓ Yes No NA Caliberation Due Date : 31-Dec-2025	
3	Fourier-Transform Infrared (FTIR) Spectrometer	Thermo Fisher Scientific		Yes ✓ No NA Caliberation Due Date :	
4	Ultraviolet-Visible (UV-Vis) Spectrophotometer	Thermo Fisher Scientific	UVVIS-07	✓ Yes No NA Caliberation Due Date : 7-June-2026	
5	Liquid Chromatography-Mass Spectrometry (LC-MS)	Waters Corporation	LSCM-09	✓ Yes No NA Caliberation Due Date : 9-Jan-2028	
6	Differential Scanning Calorimeter (DSC)	Waters Corporation	DSC-05	✓ Yes No NA Caliberation Due Date : 9-Jan-2029	
7	Nuclear Magnetic Resonance (NMR) Spectrometer	PerkinElmer		Yes ✓ No NA Caliberation Due Date :	
8	Particle Size Analyzer	PerkinElmer		Yes ✓ No NA Caliberation Due Date :	
9	Dissolution Tester	PerkinElmer	DT-005	✓ Yes No NA Caliberation Due Date : 11-June-2026	

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4. Granulation

	Recorded By:	Verified By:
1. Combine weighed materials in Glatt GPCG Pro-5. 2. Add binder solution gradually while mixing. 3. Granulate the mixture to form uniform wet mass.		
Equipment Used: GR-003		
Binder Solution Details: - Binder Solution Used: BS-05 - Volume Added: 50 mL	AS 22-Aug-23	22-Aug-23
Granulation Parameters: - Mixing Time: 63 sec - Granulation Time: 400 sec		

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7. Blending

	Recorded By:	Verified By:
1. Mix milled granules with additional excipients in V Blender. 2. Ensure uniform blend using 15 minutes of blending.		
Equipment Used: BL-003	23-Aug-23	23-Aug-2023
Blending Details: - Blending Time: 30 mins - Excipient Homogeneity: ~~3:1~~ 2.5:1		

8. Compression

	Recorded By:	Verified By:
1. Load blended mixture into Manesty F3 tablet press 2. Compress tablets at 20 kN compression force and speed of 100 tablets/minute.		
Equipment Used: _____	23-Aug-23	23-Aug-23
Compression Details: - Compression Force: 0.8N - Compression Speed: 0.3 m/s - Number of Tablets Compressed: 1024		

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5. Drying

	Recorded By:	Verified By:
1. Transfer wet mass to GPCG Pro-5. 2. Dry the granules under controlled temperature (70°C) and airflow conditions.		
Equipment Used: Air Dryer Bed-96	22-Aug-23	22-Aug-23
Drying Parameters: - Temperature: 75°C - Airflow Rate: 0.50 m3/s - Drying Time: 46 Hrs		

6. Milling

	Recorded By:	Verified By:
1. Mill dried granules using FitzMill D6A to achieve desired particle size.		
Equipment Used: ML-095	23-Aug-23	23-Aug-23
Milling Details: - Milling Time: 20 mins ① - Particle Size Achieved: 0.4 mm		

① Extra milling time as higher binder volume caused delay to achieve desired particle size

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3. Weighing and Dispensing Form

	Recorded By:	Verified By:
1. Weigh Acetaminophen and excipients accurately using Mettler Toledo XS205. 2. Record weights and lot numbers in the batch record.		
Ingredients: - Active Pharmaceutical Ingredient (API): PDKM-007 - Excipients: AL-01-05, AL-01-07		
Equipment Used: WM-02, WM-03		
Weighing Details: - API Weight: 27 mg - Excipient 1 Weight: 9 mg - Excipient 2 Weight: 10 mg	AS 22-Aug-23	22-Aug-23
Lot Numbers: - API Lot: _____ - Excipient 1 Lot: _____ - Excipient 2 Lot: _____		

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9. Coating

	Recorded By:	Verified By:
1. Transfer compressed tablets to Vector Hi-Coater. 2. Apply Opadry coating solution as per formulation requirements. 3. Dry coated tablets in controlled environment (24 hours at 25°C).		
Equipment Used: COA-213	23-Aug-23	23-Aug-23
Coating Details: - Coating Solution Used: PDKM-005-COA - Coating Time: 22 mins - Coating Thickness Achieved: 95 um		